

MASTER APTITUDE PREP GUIDE

Ultra Source — All Topics, Sub-Topics, Formulas, Key Points, Repeated PYQs with Solutions

For Tech & Non-Tech Company Placement / Recruitment Aptitude Tests

Covers: Quantitative Aptitude | Logical Reasoning | Verbal Ability | Data Interpretation

Company Patterns: TCS, Infosys, Wipro, Accenture, Cognizant, Capgemini, HCL, IBM, SSC, Banking (IBPS/SBI), Insurance

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PART A — Quantitative Aptitude

A1. Number System

Sub-topics: Classification, Divisibility, Unit Digit, Factors, Remainders

Natural (1,2,3..) → Whole (0,1,2..) → Integers (...,-1,0,1,...) → Rational → Irrational → Real → Complex

Divisible by	Rule
2	Last digit even (0,2,4,6,8)
3	Sum of digits divisible by 3
4	Last 2 digits divisible by 4
5	Last digit 0 or 5
6	Divisible by 2 and 3
7	Double last digit, subtract from rest; result divisible by 7
8	Last 3 digits divisible by 8
9	Sum of digits divisible by 9
10	Last digit 0
11	(Sum of digits at odd places) – (sum at even places) = 0 or multiple of 11
12	Divisible by 3 and 4
25	Last 2 digits divisible by 25 (00,25,50,75)

Key Point: Unit digit cyclicity — 2:(2,4,8,6) cycle 4 | 3:(3,9,7,1) cycle 4 | 4:(4,6) cycle 2 | 7:(7,9,3,1) cycle 4 | 8:(8,4,2,6) cycle 4 | 9:(9,1) cycle 2 | 0,1,5,6 always give themselves. To find unit digit of a^b : find $(b \bmod \text{cycle-length})$, map to cycle (use cycle length if remainder 0).

Sum of first n natural numbers = $\frac{n(n+1)}{2}$ | **Sum of squares** = $\frac{n(n+1)(2n+1)}{6}$ | **Sum of cubes** = $[\frac{n(n+1)}{2}]^2$

Number of factors of $N = p_1^{a_1} \times p_2^{a_2} \dots \rightarrow (a_1+1)(a_2+1) \dots$

Sum of factors = $[(\frac{p_1^{a_1+1}-1}{p_1-1}) \times (\frac{p_2^{a_2+1}-1}{p_2-1})] \dots$

Q1 (PYQ - TCS/Infosys repeated). Find the unit digit of 7^{95} .

Solution: Cycle of 7 is (7,9,3,1), length 4. $95 \bmod 4 = 3 \rightarrow$ 3rd term in cycle = 3. **Answer: 3**

Q2. A number when divided by 342 gives remainder 47. What is the remainder when the same number is divided by 18?

Solution: $342 = 18 \times 19$, so number = $342k + 47$. $47 \bmod 18 = 11$. **Answer: 11** (Rule: if divisor divides the bigger divisor exactly, remainder = original remainder mod new divisor.)

A2. HCF & LCM

HCF $\times$ LCM = Product of two numbers (only for 2 numbers)

HCF of fractions = $\frac{\text{HCF}(\text{numerators})}{\text{LCM}(\text{denominators})}$ | **LCM of fractions** = $\frac{\text{LCM}(\text{numerators})}{\text{HCF}(\text{denominators})}$

Key Point: Largest number that divides x, y, z leaving remainders $a, b, c \rightarrow \text{HCF}(x-a, y-b, z-c)$. Largest number dividing x, y, z leaving the *same* remainder $\rightarrow \text{HCF of differences } (x-y, y-z, x-z)$. Smallest number divisible by x, y, z leaving remainder r in each case $\rightarrow \text{LCM}(x, y, z) + r$.

Q3 (PYQ - Wipro). Find the greatest number that divides 245 and 1029 leaving remainder 5 in each case.

Solution: Subtract remainder: 240 and 1024. $HCF(240, 1024)$: $240 = 2^4 \times 3 \times 5$, $1024 = 2^{10} \rightarrow HCF = 16$. **Answer: 16**

A3. Percentages

$x\%$ of $y = (x/100) \times y$ | $\% \text{ change} = (\text{Change}/\text{Original}) \times 100$

If A is R% more than B, B is less than A by $[R/(100+R)] \times 100\%$. If A is R% less than B, B is more than A by $[R/(100-R)] \times 100\%$.

Fraction-% table (memorize): $1/2 = 50\%$, $1/3 = 33.33\%$, $1/4 = 25\%$, $1/5 = 20\%$, $1/6 = 16.67\%$, $1/7 = 14.28\%$, $1/8 = 12.5\%$, $1/9 = 11.11\%$, $1/10 = 10\%$, $1/11 = 9.09\%$, $1/12 = 8.33\%$, $1/16 = 6.25\%$, $1/20 = 5\%$, $1/25 = 4\%$.

Successive % change a%, b%: net change = $a + b + (ab/100)\%$

Q4 (PYQ - Accenture). The price of an item is increased by 25% and then decreased by 20%. Net % change?

Solution: Net = $25 - 20 + (25 \times -20/100) = 5 - 5 = 0\%$. **Answer: No change**

Q5. In an election, a candidate got 65% of votes and won by 5400 votes. Total votes?

Solution: Winning margin % = $65 - 35 = 30\%$. 30% of total = 5400 $\rightarrow$ Total = $5400/0.30 = 18000$. **Answer: 18000**

A4. Profit, Loss & Discount

$\text{Profit}\% = (SP - CP)/CP \times 100$ | $\text{Loss}\% = (CP - SP)/CP \times 100$

$SP = CP(1 + P\%/100)$ or $CP(1 - L\%/100)$ | **Marked Price after discount d%: $SP = MP(1 - d/100)$**

Key Point: If cost price of x articles = selling price of y articles, $\text{Profit}\% = [(x-y)/y] \times 100$. Successive discounts $d_1\%$, $d_2\%$ $\rightarrow$ net discount = $d_1 + d_2 - (d_1 d_2/100)$.

Q6 (PYQ - Capgemini). A shopkeeper marks goods 40% above cost and allows 25% discount. Find profit%.

Solution: Let $CP = 100$, $MP = 140$, $SP = 140 \times 0.75 = 105$. $\text{Profit}\% = 5\%$. **Answer: 5%**

A5. Simple Interest & Compound Interest

$SI = (P \times R \times T)/100$ | $A = P + SI$

$CI = P[(1 + R/100)^T - 1]$ | **Amount = $P(1 + R/100)^T$**

CI compounded half-yearly: $A = P(1 + (R/2)/100)^{(2T)}$ | Quarterly: $A = P(1 + (R/4)/100)^{(4T)}$

Key Point: Difference between CI and SI for 2 years = $P(R/100)^2$. For 3 years: $\text{Diff} = P(R/100)^2 \times [(300+R)/100]$. Rate of interest doubling money in T years (SI) = $100/T$.

Q7 (PYQ - IBPS). The difference between CI and SI on a sum for 2 years at 10% p.a. is Rs.150. Find the sum.

Solution: $\text{Diff} = P(R/100)^2 \rightarrow 150 = P \times (0.1)^2 = P \times 0.01 \rightarrow P = 15000$. **Answer: Rs. 15,000**

A6. Ratio, Proportion & Variation

$a:b = a/b$. If $a:b = b:c$, b is mean proportional, $b^2 = ac$. Fourth proportional to a, b, c = bc/a .

Key Point: If $A:B = a:b$ and $B:C = c:d$, then $A:B:C = ac : bc : bd$. Direct variation: $x/y=k$ (constant). Inverse variation: $xy=k$.

Q8. Rs.1620 divided among A,B,C in ratio 2:3:4. Find C's share.

Solution: Total parts=9, $C = (4/9) \times 1620 = 720$. **Answer: Rs. 720**

A7. Averages

Average = Sum of observations / Number of observations

Key Point: If one value in a group of n is replaced by a new value, change in average = $(\text{new value} - \text{old value})/n$.
Average speed for equal distances at speeds a & $b = 2ab/(a+b)$.

Q9 (PYQ - Cognizant). Average age of 30 students is 12 years. If teacher's age is included, average becomes 13. Teacher's age?

Solution: Total age(students)=360. New total for 31 = $13 \times 31 = 403$. Teacher's age = $403 - 360 = 43$. **Answer: 43 years**

A8. Problems on Ages

Key Point: Represent present ages as variables, translate "years ago/hence" carefully as $(\text{age} \pm \text{years})$. Form 2 equations and solve simultaneously.

Q10. Father is 3 times as old as son. After 12 years father will be twice as old as son. Find present ages.

Solution: $F=3S$. $F+12=2(S+12) \rightarrow 3S+12=2S+24 \rightarrow S=12$, $F=36$. **Answer: Son 12 yrs, Father 36 yrs**

A9. Time & Work (+ Pipes & Cisterns)

If A does work in x days, A's 1-day work = $1/x$. Combined 1-day work = $1/x + 1/y \rightarrow \text{days} = xy/(x+y)$

Pipes: Inlet pipe +ve rate, outlet pipe -ve rate. Time to fill = $1/(\text{net rate})$

Key Point: If A is k times as efficient as B, ratio of time taken $A:B = 1:k$. Work & wages: wages divided in ratio of work done (= ratio of (efficiency $\times$ time)).

Q11 (PYQ - TCS NQT). A can do a work in 12 days, B in 15 days. They work together for 4 days, then A leaves. In how many more days will B finish?

Solution: Combined rate/day = $1/12 + 1/15 = 9/60 = 3/20$. Work in 4 days = $12/20 = 3/5$. Remaining = $2/5$. B's rate = $1/15$. Days = $(2/5)/(1/15) = 6$. **Answer: 6 days**

A10. Time, Speed & Distance (+ Trains, Boats & Streams)

Speed = Distance/Time | km/hr to m/s: $\times 5/18$ | m/s to km/hr: $\times 18/5$

Trains: Time to cross a pole/point = Length of train / Speed. Time to cross a platform = (Length of train + Length of platform)/Speed. Two trains crossing (opposite direction): relative speed = sum of speeds; (same direction): relative speed = difference of speeds.

Boats & Streams: Downstream speed = $u+v$, Upstream speed = $u-v$, where u =boat speed, v =stream speed. Boat speed = $\frac{1}{2}(\text{down} + \text{up})$, Stream speed = $\frac{1}{2}(\text{down} - \text{up})$

Key Point: If a man changes his speed in ratio $a:b$, time taken is in ratio $b:a$ (inverse). Average speed (equal time, not equal distance) = simple average of speeds.

Q12 (PYQ - Infosys). Two trains of length 120m and 150m run in opposite directions on parallel tracks at 60 km/h and 40 km/h. Time to cross each other?

Solution: Relative speed = 100 km/h = $100 \times \frac{5}{18} = \frac{250}{9}$ m/s. Total length = 270m. Time = $270 / (\frac{250}{9}) = 9.72s \approx 9.7$ seconds

A11. Mixtures & Alligations

Alligation rule: $(\text{Quantity of cheaper}) / (\text{Quantity of dearer}) = (\text{CP of dearer} - \text{Mean price}) / (\text{Mean price} - \text{CP of cheaper})$

Q13. In what ratio must rice at Rs.60/kg be mixed with rice at Rs.75/kg so that mixture costs Rs.65/kg?

Solution: Cheaper=60, Dearer=75, Mean=65. Ratio = $(75-65):(65-60) = 10:5 = 2:1$. **Answer: 2:1**

A12. Permutations & Combinations

$nPr = \frac{n!}{(n-r)!}$ | $nCr = \frac{n!}{r!(n-r)!}$ | $nCr = nC(n-r)$

Key Point: Circular permutations of n objects = $(n-1)!$. If clockwise/anticlockwise same, $(n-1)!/2$. Word arrangement with repeated letters: $\frac{n!}{(p!q!...)}$ where p,q are counts of repeated letters.

Q14 (PYQ - Wipro). In how many ways can the letters of the word "MISSISSIPPI" be arranged?

Solution: 11 letters: M-1, I-4, S-4, P-2. Arrangements = $\frac{11!}{(4!4!2!)} = 34,650$. **Answer: 34,650**

A13. Probability

$P(E) = \text{Favourable outcomes} / \text{Total outcomes}$. $0 \leq P(E) \leq 1$. $P(\text{not } E) = 1 - P(E)$

Key Point: "AND" (independent events) → multiply probabilities. "OR" (mutually exclusive) → add probabilities. Deck of cards: 52 total, 4 suits×13, 26 red/26 black, 12 face cards, 4 aces.

Q15. Two dice are thrown. Probability that sum is 8?

Solution: Total outcomes=36. Favourable (sum=8): (2,6)(3,5)(4,4)(5,3)(6,2)=5. $P=5/36$. **Answer: 5/36**

A14. Algebra (Linear/Quadratic Equations, AP/GP)

Quadratic $ax^2+bx+c=0$: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$. Sum of roots = $-b/a$, Product = c/a

AP: nth term = $a+(n-1)d$. Sum = $\frac{n}{2}[2a+(n-1)d]$. **GP:** nth term = $ar^{(n-1)}$. Sum (finite) = $\frac{a(r^n-1)}{(r-1)}$; Sum (infinite, $|r|<1$) = $\frac{a}{(1-r)}$

$(a+b)^2=a^2+2ab+b^2$ | $(a-b)^2=a^2-2ab+b^2$ | $a^2-b^2=(a+b)(a-b)$ | $(a+b)^3=a^3+b^3+3ab(a+b)$ | $a^3+b^3=(a+b)(a^2-ab+b^2)$

Q16 (PYQ - Accenture). Find the sum of first 20 terms of AP: 3, 7, 11, 15...

Solution: $a=3$, $d=4$, $n=20$. Sum = $\frac{20}{2}[2(3)+19(4)] = 10[6+76]=820$. **Answer: 820**

A15. Geometry & Mensuration

Shape	Area	Perimeter / Surface Area
Square	a^2	Perimeter $4a$; Diagonal $a\sqrt{2}$

Rectangle	$l \times b$	Perimeter $2(l+b)$
Triangle	$\frac{1}{2} \times \text{base} \times \text{height}$; $\sqrt{s(s-a)(s-b)(s-c)}$ (Heron's)	Perimeter $a+b+c$
Equilateral Triangle	$(\sqrt{3}/4)a^2$	Perimeter $3a$
Circle	πr^2	Circumference $2\pi r$
Parallelogram	$\text{base} \times \text{height}$	—
Trapezium	$\frac{1}{2}(a+b) \times h$	—
Rhombus	$\frac{1}{2} \times d_1 \times d_2$	Perimeter $4a$
Cube	Vol a^3	Surface area $6a^2$; Diagonal $a\sqrt{3}$
Cuboid	Vol $l \times b \times h$	Surface area $2(lb+bh+lh)$
Cylinder	Vol $\pi r^2 h$	CSA $2\pi rh$; TSA $2\pi r(r+h)$
Cone	Vol $(1/3)\pi r^2 h$	CSA πrl (l =slant height); TSA $\pi r(r+l)$
Sphere	Vol $(4/3)\pi r^3$	Surface area $4\pi r^2$
Hemisphere	Vol $(2/3)\pi r^3$	CSA $2\pi r^2$; TSA $3\pi r^2$

Key Point: Pythagoras: $h^2 = p^2 + b^2$. Pythagorean triplets: (3,4,5), (5,12,13), (8,15,17), (7,24,25), (20,21,29). Angle sum of polygon = $(n-2) \times 180^\circ$.

Q17 (PYQ - HCL). Find the volume of a cone with radius 7cm and height 24cm. ($\pi=22/7$)

Solution: Vol = $(1/3) \times (22/7) \times 49 \times 24 = (1/3) \times 22 \times 7 \times 24 = 1232 \text{ cm}^3$. **Answer: 1232 cm³**

A16. Data Interpretation

Sub-topics: Tables, Bar Graphs, Line Graphs, Pie Charts, Caselets, Mixed Graphs

Key Point: Pie chart: each 1% = 3.6° . Always check the "base" (total) before computing %. In DI sets, calculate ratios and % changes carefully; approximate first to eliminate options, then compute exactly if needed. Common trap: reading "increase" vs "increase in %" vs "% of total" — read the question twice.

A17. Data Sufficiency (Quant)

Key Point: Standard answer choices: (A) Statement I alone sufficient, (B) Statement II alone sufficient, (C) Both together sufficient but neither alone, (D) Either alone sufficient, (E) Neither sufficient. Approach: test Statement I alone first, forget II; then test II alone forgetting I; only then combine.

A18. Simplification, BODMAS & Surds/Indices

BODMAS: Brackets → Of → Division → Multiplication → Addition → Subtraction

$a^m \times a^n = a^{(m+n)}$ | $a^m / a^n = a^{(m-n)}$ | $(a^m)^n = a^{(mn)}$ | $a^0 = 1$ | $a^{-n} = 1/a^n$ | $a^{(1/n)} = \text{nth root of } a$

A19. Partnership

Key Point: Profit sharing ratio = ratio of (capital × time invested) for each partner, when time periods differ.

Q18. A invests Rs.5000 for 12 months, B invests Rs.6000 for 8 months. Profit ratio?

Solution: A:B = $5000 \times 12 : 6000 \times 8 = 60000 : 48000 = 5:4$. **Answer: 5:4**

A20. Logarithms

$\log(mn) = \log m + \log n$ | $\log(m/n) = \log m - \log n$ | $\log(m^n) = n \cdot \log m$ | $\log_a a = 1$ | $\log_a 1 = 0$ | Change of base:
 $\log_a b = \log b / \log a$

PART B — Logical Reasoning

B1. Number, Letter & Alpha-Numeric Series

Key Point: Check differences (arithmetic), ratios (geometric), squares/cubes, prime series, or alternating patterns. For letter series convert to numbers: A=1...Z=26.

Q19 (PYQ - TCS). Find the next term: 2, 6, 12, 20, 30, ?

Solution: Differences: 4,6,8,10,12 ($n(n+1)$ pattern). Next = $30+12=42$. **Answer: 42**

B2. Coding-Decoding

Key Point: Common types: Letter shifting (+1/-1 etc.), word coding, number coding, substitution coding. Always find the logic on the given example then apply to the question — don't assume standard alphabet shift without checking.

Q20. If CAT is coded as 24-2-40, how is DOG coded (same logic: letter position $\times 2$)?

Solution: C=3 $\rightarrow$ 6? but given 24 $\rightarrow$ check: alphabet position reversed (C=24th from end since Z=1)? Actually here logic = (position from start) $\times 2$: C=3 $\times 2=6$ not 24 — recheck: position from end: C is 3rd from start, 24th from end ($26-3+1=24$). A=1st from start=26th from end $\rightarrow$ but coded as 2. Actual logic: position from start $\times 2$: A=1 $\times 2=2$ ✓, C=3 $\times 2=6$ (not 24, so recheck T): T=20 $\times 2=40$ ✓. So C should be 6, meaning likely a typo in classic version — key takeaway: **always verify pattern on ALL given letters before applying**. Using verified rule (position $\times 2$): D=4 $\times 2=8$, O=15 $\times 2=30$, G=7 $\times 2=14$. **Answer: 8-30-14**

B3. Blood Relations

Key Point: Draw a family tree diagram; use '+' for male, '-' for female if gender given. Common relations to memorize: Mother's/Father's brother = Uncle; Uncle's son = Cousin; Son's wife = Daughter-in-law; Husband's/wife's sister = Sister-in-law; Sister's husband = Brother-in-law.

Q21 (PYQ - IBPS). Pointing to a photo, a man says "She is the daughter of my grandfather's only son." How is she related to the man?

Solution: Grandfather's only son = the man's father (or the man himself if no other siblings — standard answer assumes father). Daughter of father = Sister. **Answer: Sister**

B4. Direction Sense Test

Key Point: Standard compass: facing North, right turn=East, left turn=West. Use coordinate method (x,y) — moving North=+y, South=-y, East=+x, West=-x — then use distance formula $\sqrt{x^2+y^2}$ for shortest distance.

Q22. A man walks 5km North, then 3km East, then 5km South. How far is he from start and in which direction?

Solution: Net movement: North $5-5=0$, East=3. He is 3km East of start. **Answer: 3 km, East**

B5. Syllogisms

Key Point: Use Venn diagrams. "All A are B" = A circle inside B. "No A is B" = separate circles. "Some A are B" = overlapping circles. Consider ALL possible diagram cases before concluding — a conclusion is valid only if true in every possible case.

Q23. Statements: All pens are books. Some books are pencils. Conclusion: I. Some pencils are pens. II. Some books are pens.

Solution: "All pens are books" → automatically gives "Some books are pens" (conversion of universal affirmative) — Conclusion II follows. Conclusion I doesn't necessarily follow (no direct link established between pens & pencils). **Answer: Only conclusion II follows**

B6. Seating Arrangement (Linear & Circular)

Key Point: For circular arrangement facing centre, clockwise = to the right; facing outward reverses left/right. Draw the diagram, place fixed/most-constrained clues first, then fill by elimination. In linear rows, note "facing north" vs "facing south" flips left-right.

B7. Puzzles (Box / Floor / Linear ranking)

Key Point: Make a grid/table of variables (name × attribute). Process direct clues first, then negative/indirect clues, use elimination. Floor puzzles: "lives above/below" — number floors bottom to top unless stated otherwise.

B8. Data Sufficiency (Reasoning)

Key Point: Same 5-option method as quant DS. Common trap: assuming real-world facts not stated in the statements.

B9. Statement-Conclusion / Statement-Assumption / Course of Action

Key Point: Assumption = something implicit taken for granted (must be necessary for the statement to make sense). Conclusion must follow logically, not just seem "likely true." Course of action must be practical, relevant, and address the problem — not a mere suggestion to "study further" unless explicitly reasonable.

B10. Venn Diagrams (Numerical / Set-based)

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) \quad | \quad n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

Q24 (PYQ - Cognizant). In a class of 60, 25 play cricket, 20 play football, 10 play both. How many play neither?

Solution: Play at least one = $25 + 20 - 10 = 35$. Neither = $60 - 35 = 25$. **Answer: 25**

B11. Clocks & Calendars

Angle between hands at H:M = $|(30 \times H) - (11/2 \times M)|$ degrees. Hands overlap every $65 \frac{1}{11}$ minutes. Minute hand gains 5.5° per minute over hour hand.

Calendar: Odd days per 100 years=5, 200 yrs=3, 300 yrs=1, 400 yrs=0. Leap year: divisible by 4, but century years only if divisible by 400.

Q25. Find the angle between hour and minute hands at 3:40.

Solution: Angle = $|30 \times 3 - 11/2 \times 40| = |90 - 220| = 130^\circ$. **Answer: 130°**

B12. Non-Verbal Reasoning

Sub-topics: Mirror images, water images, paper folding/cutting, embedded figures, figure series, odd-

one-out figures, completion of figure

Key Point: Mirror image reverses left-right (flips horizontally along a vertical mirror line). Water image reverses up-down (flips vertically) and inverts the figure. These are typically visual/diagram-based questions — practice with actual figure sets in a dedicated reasoning book/app since they can't be fully represented in text.

B13. Cubes & Dice

Key Point: Standard dice: opposite faces sum patterns vary by dice type — for a standard dice, if two positions of the same dice are given, use the "sharing an edge" rule to deduce opposite faces (any two faces NOT appearing together across the given views are opposite). For cube-cutting problems (a big cube cut into n^3 smaller cubes and painted): cubes with 3 faces painted = 8 (corners), 2 faces = $12(n-2)$, 1 face = $6(n-2)^2$, 0 faces = $(n-2)^3$.

Q26 (PYQ - Wipro). A cube is painted on all faces and cut into 64 smaller equal cubes. How many have exactly 2 faces painted?

Solution: $64=4^3 \rightarrow n=4$. 2-face painted = $12(n-2)=12 \times 2=24$. **Answer: 24**

B14. Analogy, Classification & Odd One Out

Key Point: Identify the relationship type first (synonym, antonym, part-whole, cause-effect, function, category) in the given pair, then apply the exact same relationship to find the answer.

PART C — Verbal Ability

C1. Reading Comprehension

Key Point: Read the question stems before the passage to know what to look for. Identify passage tone/author's attitude, main idea (usually 1st & last paragraph), and locate specifics by keyword scanning rather than re-reading the whole passage. Watch for "EXCEPT / NOT true" questions — one wrong twist can flip the answer.

C2. Synonyms & Antonyms

Key Point: Learn words in context/root-word groups rather than isolated lists. Common prefixes: "ben-/bene-" = good, "mal-" = bad, "un-/in-/dis-" = negation, "pro-" = forward/favor, "anti-" = against. Watch for "connotation" traps — a word can be a near-synonym but wrong shade of meaning.

C3. Sentence Correction / Grammar

Common rules tested:

- Subject-verb agreement (singular subject → singular verb, even with intervening phrases)
- Parallelism in lists ("...enjoys reading, writing, and to swim" is wrong — must be "swimming")
- Correct tense sequencing (past perfect for the earlier of two past events)
- Pronoun-antecedent agreement (number and gender must match)
- Preposition usage: "different from" (not "than," in formal usage), "married to", "capable of"
- Articles: "a" before consonant sound, "an" before vowel sound (an hour, a university)

C4. Para Jumbles (Sentence Rearrangement)

Key Point: Find the opening sentence (introduces topic generally, no pronouns referring back). Look for linking words (however, therefore, this, thus) to chain sentences. The sentence with a pronoun ("it", "this", "they") usually cannot be first.

C5. Fill in the Blanks / Cloze Test

Key Point: Read the full sentence/passage first for context/tone before choosing. Eliminate options that are grammatically wrong even if the meaning seems to fit.

C6. One Word Substitution

Phrase	One Word
One who loves books	Bibliophile
Study of birds	Ornithology
Fear of heights	Acrophobia
Government by the people	Democracy
One who does not believe in God	Atheist
A place where orphans live	Orphanage
Handwriting that cannot be read	Illegible
One who knows many languages	Polyglot

C7. Idioms & Phrases

Idiom	Meaning
Once in a blue moon	Very rarely
Break the ice	Start a conversation / ease tension
Bite the bullet	Face a difficult situation bravely
Piece of cake	Something very easy
Under the weather	Feeling unwell
Cost an arm and a leg	Very expensive

PART D — Company-wise Test Patterns

Patterns change periodically — always confirm the latest pattern from the official company careers page/latest test-experience posts before your exam. Below reflects commonly reported recent structures.

D1. Tech / IT Services Companies

Company	Sections	Typical Focus
TCS NQT	Numerical Ability, Reasoning, Verbal, Programming Logic (MCQ), optional coding	Foundation section is qualifying; advanced section for higher roles
Infosys (InfyTQ / SP)	Quant, Reasoning, Verbal, Pseudocode, Coding	Reasoning-heavy, includes puzzles
Wipro (WILP/Elite)	Quant, Logical, Verbal, Essay writing, Coding	Includes written English essay
Accenture	Cognitive (Quant, Verbal, Reasoning), Coding, Communication (voice/text)	Includes coding assessment + communication round
Cognizant (GenC)	Aptitude (Quant, Reasoning, Verbal), Automata (coding)	Automata coding round is key differentiator
Capgemini	Game-based Pymetrics-style, Quant, Logical, Verbal, Coding	Includes behavioral/gamified assessment
HCL	Quant, Reasoning, Verbal, optional Technical/Coding	Shorter, moderate difficulty
IBM	Quant, Reasoning, Verbal, Technical MCQs	Includes domain technical MCQs

D2. Non-Tech / Government / Banking / Other Sector Exams

Exam/Sector	Sections	Typical Focus
IBPS PO/Clerk, SBI PO/Clerk	Quant, Reasoning, English, General/Banking Awareness, Computer Aptitude	Heavy DI and Puzzle-based reasoning
SSC CGL/CHSL	Quant, Reasoning, English, General Awareness	General knowledge weightage is significant
Insurance (LIC/NIACL)	Reasoning, Quant, English, General Awareness	Similar to banking pattern
Railways (RRB)	Maths, General Intelligence & Reasoning, General Science, General Awareness	Includes basic science section
MBA Entrance (CAT/MAT/CMAT)	Quant (DILR), Verbal & RC, Data Interpretation & Logical Reasoning	Higher difficulty, time-pressure heavy
Corporate Non-Tech (Sales/Ops/HR generalist hiring)	Quant, Reasoning, Verbal, Situational Judgement	Situational judgement tests workplace behavior

Key Point: Across almost every company, Quant + Reasoning + Verbal is the common core. Tech companies add a Coding/Programming-Logic section; Banking/Govt exams add General/Banking Awareness and heavier Data Interpretation & Puzzles; MBA exams add Reading Comprehension weightage and tighter sectional timing.

PART E — Master Formula Sheet (Quick Revision)

Topic	Formula
Percentage	$\% \text{change} = (\text{Change}/\text{Original}) \times 100$
Profit/Loss	$P\% = (\text{SP} - \text{CP})/\text{CP} \times 100$
SI	$(P \times R \times T)/100$
CI	$P[(1 + R/100)^T - 1]$
Avg. Speed (equal dist.)	$2ab/(a+b)$
Time & Work	Combined rate = $1/x + 1/y$
Alligation	Cheaper qty : Dearer qty = (Dearer – Mean):(Mean – Cheaper)
nPr	$n!/(n-r)!$
nCr	$n!/[r!(n-r)!]$
Probability	Favourable/Total
AP sum	$n/2[2a + (n-1)d]$
GP sum (infinite)	$a/(1-r), r < 1$
Circle	Area πr^2 ; Circumference $2\pi r$
Cylinder	Vol $\pi r^2 h$; TSA $2\pi r(r+h)$
Cone	Vol $(1/3)\pi r^2 h$; CSA $\pi r l$
Sphere	Vol $(4/3)\pi r^3$; SA $4\pi r^2$
Clock angle	$ 30H - 5.5M $
Set theory (2 sets)	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
HCF $\times$ LCM	= product of two numbers
km/hr $\leftrightarrow$ m/s	$\times 5/18$ and $\times 18/5$

PART F — Additional Top Repeated PYQs with Solutions

Q27 (PYQ - TCS, very frequently repeated). A train 150m long is running at 68 km/hr. In what time will it pass a man running at 8 km/hr in the same direction?

Solution: Relative speed = $68 - 8 = 60$ km/hr = $60 \times 5/18 = 50/3$ m/s. Time = $150 / (50/3) = 9$ sec. **Answer: 9 seconds**

Q28 (PYQ - Infosys). If 20 men can build a wall in 15 days, how many men are needed to build it in 10 days?

Solution: Men $\times$ Days = constant = $20 \times 15 = 300$. Men needed = $300 / 10 = 30$. **Answer: 30 men**

Q29 (PYQ - Wipro). A sum of money doubles itself in 8 years at simple interest. In how many years will it triple?

Solution: Doubles in 8 yrs $\rightarrow$ SI in 8 yrs = P. Rate = $(100 \times P) / (P \times 8) = 12.5\%$. To triple, SI needed = $2P$. Time = $(2P \times 100) / (P \times 12.5) = 16$ years. **Answer: 16 years**

Q30 (PYQ - Accenture). The average of 5 consecutive odd numbers is 61. Find the largest number.

Solution: For consecutive numbers average = middle number = 61. Numbers: 57, 59, 61, 63, 65. Largest = 65. **Answer: 65**

Q31 (PYQ - Cognizant). A can complete a work in 20 days and B in 30 days. They started together, but A left 5 days before completion. Total days taken?

Solution: Let total days = x. B works all x days, A works (x-5) days. $(x-5)/20 + x/30 = 1$. Multiply by 60: $3(x-5) + 2x = 60 \rightarrow 3x - 15 + 2x = 60 \rightarrow 5x = 75 \rightarrow x = 15$. **Answer: 15 days**

Q32 (PYQ - Capgemini). A boat covers 32 km upstream in 4 hours and 36 km downstream in 3 hours. Find speed of boat in still water.

Solution: Upstream speed = $32/4 = 8$ km/hr. Downstream speed = $36/3 = 12$ km/hr. Boat speed = $\frac{1}{2}(12 + 8) = 10$ km/hr. **Answer: 10 km/hr**

Q33 (PYQ - HCL). What is the compound interest on Rs.10,000 for 2 years at 10% per annum, compounded annually?

Solution: A = $10000 \times (1.1)^2 = 12100$. CI = $12100 - 10000 = 2100$. **Answer: Rs. 2100**

Q34 (PYQ - IBM). Ravi's salary was first increased by 20% and then decreased by 20%. Find the net change.

Solution: Net% = $20 - 20 + (20 \times -20/100) = 0 - 4 = -4\%$. **Answer: 4% decrease**

Q35 (PYQ - SSC/Banking, reasoning). Pointing to a woman, Ravi said, "Her mother is the only daughter of my mother." How is Ravi related to the woman?

Solution: Only daughter of Ravi's mother = Ravi's sister = the woman's mother. So Ravi is the woman's maternal uncle (mama). **Answer: Maternal Uncle**

Q36 (PYQ - IBPS, DI). If the ratio of boys to girls in a class of 60 is 3:2, and 20% of boys and 25% of girls are absent, how many students are present?

Solution: Boys = 36, Girls = 24. Absent boys = $20\% \times 36 = 7.2$ (round per context) present boys = 28.8; typically framed with round numbers — using exact ratio math: Present = $36 \times 0.8 + 24 \times 0.75 = 28.8 + 18 = 46.8 \approx 47$ students (values chosen for clean numbers in real papers; verify given numbers divide evenly)

Q37 (PYQ - common across companies). Find the missing number: 5, 11, 19, 29, 41, ?

Solution: Differences: 6, 8, 10, 12, 14 (increasing by 2). Next = $41 + 14 = 55$. **Answer: 55**

Q38 (PYQ - Verbal/Cloze). Choose the correct word: "Despite ___ hard, he failed the exam." (a) work (b) working (c) worked (d) works

Solution: "Despite" is followed by a gerund/noun form. **Answer: (b) working**

Q39 (PYQ - Geometry, TCS). The area of a square is equal to the area of a rectangle whose length is 25cm and breadth is 16cm. Find the perimeter of the square.

Solution: Area = $25 \times 16 = 400 = \text{side}^2 \rightarrow \text{side} = 20$. Perimeter = $4 \times 20 = 80\text{cm}$. **Answer: 80 cm**

Q40 (PYQ - Number series/Coding, Accenture). In a certain code, "TIGER" is written as "UJHFS". How is "LION" written in the same code?

Solution: Each letter shifted +1 (T→U, I→J, G→H, E→F, R→S). Apply to LION: L→M, I→J, O→P, N→O. **Answer: MJPO**

PART G — Facts, Traps & Things to Remember

G1. High-Frequency Facts to Memorize

- 1 to 20 squares and 1 to 15 cubes should be memorized instantly (square of 17=289, 18=324, 19=361; cube of 12=1728, 13=2197).
- Prime numbers up to 100: 2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97 (25 primes total).
- Standard fraction-to-% conversions (see A3 table) — used in almost every % / DI question.
- $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{5} \approx 2.236$ — needed for geometry/mensuration shortcuts.
- $\pi \approx 22/7 \approx 3.14159$.
- 100 years = 5 odd days; leap year has 2 odd days, normal year has 1 odd day (calendar problems).

G2. Common Traps & Mistakes

- **% increase/decrease base confusion:** "increased by 20%" uses the original value as base, not the new value — a very common trap in reused/repeated questions.
- **CP vs SP mix-up** in profit/loss — always fix CP=100 as a base for quick calculation.
- **"Per annum" vs total interest** — check if rate given is annual even when compounding is half-yearly/quarterly (adjust rate and time accordingly).
- **Relative speed direction** — same direction subtract, opposite direction add (train/boat problems).
- **Syllogism "possibility" cases** — a conclusion using "possibility" is different from "definitely true"; many students wrongly reject possibility-based options.
- **Reading DI/RC too fast** — misreading "at least" vs "at most", or "increase" vs "increase in %" leads to wrong answers even with correct calculation.
- **Unit conversion errors** — km/hr $\leftrightarrow$ m/s, or $\text{cm}^3 \leftrightarrow$ litres (1000 cm^3 = 1 litre) are frequently tested silently within word problems.
- **Negative marking** — many company tests (and all banking/SSC exams) have negative marking; don't guess blindly on totally unknown questions.

G3. Quick Mental-Math Shortcuts

- To multiply by 5: multiply by 10, divide by 2. To multiply by 25: multiply by 100, divide by 4. To multiply by 9: multiply by 10, subtract the number.
- Squaring numbers near 50: $(50 \pm x)^2 = 2500 \pm 100x + x^2$. e.g., $53^2 = 2500 + 300 + 9 = 2809$.
- Multiplying two numbers near a base (e.g., near 100): use $(100+a)(100+b) = 10000 + 100(a+b) + ab$.
- To check divisibility/factor quickly during MCQ elimination, back-substitute answer options into the equation rather than solving algebraically from scratch.

PART H — Preparation Strategy & Time Management

H1. Suggested Study Sequence

1. **Week 1-2:** Build formula base — Number System, Percentages, Profit/Loss, SI/CI, Ratio/Averages (foundation arithmetic used everywhere else).
2. **Week 3-4:** Time-Speed-Distance, Time & Work, Permutation-Combination, Probability, Algebra, Mensuration.
3. **Week 5:** Full Logical Reasoning topics (series, coding, blood relations, arrangements, puzzles, syllogisms).
4. **Week 6:** Verbal Ability + Reading Comprehension + Data Interpretation practice sets.
5. **Week 7 onward:** Full-length mock tests under real time limits, review every wrong/guessed answer, maintain an error log.

H2. Exam-Day Time Management

- Do a first pass and answer only questions you can solve in under 45-60 seconds; mark and skip the rest.
- In sectional-timed tests, never spend more than 20% of extra time on a single tough question — lost time can't be recovered.
- For DI/RC sets, skim the data/passage once fully before attempting the linked questions — saves re-reading time.
- Attempt easier sections first if the platform allows switching, to lock in guaranteed marks before tackling harder ones.
- Always keep the last 5 minutes to revisit marked/uncertain questions rather than starting a fresh difficult one.

H3. Final Checklist Before the Test

- Revise the Master Formula Sheet (Part E) the night before / morning of the test.
- Know the exact test pattern, number of sections, sectional time limits and negative marking rules for your specific company (Part D).
- Practice at least 2-3 full-length mock tests in the same time-bound, no-pause format as the real test.
- Keep a personal error log of mistakes from mocks and review it right before the exam — repeated PYQs recur across drives from the same company/vendor.

— End of Master Aptitude Prep Guide — All the best for your placement drive! —